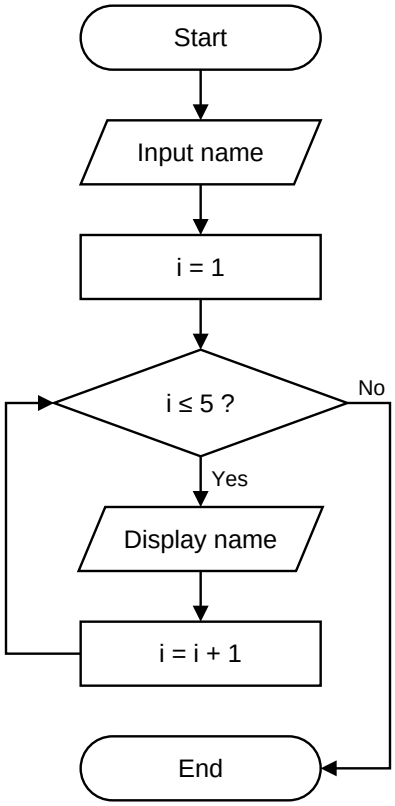
1 Print Your Name

Repetition structure: FOR

Input a name and display it five times.



Simulation table

Input: name = “Maria”. Each row = one check of i ≤ 5; shaded row = loop ends.

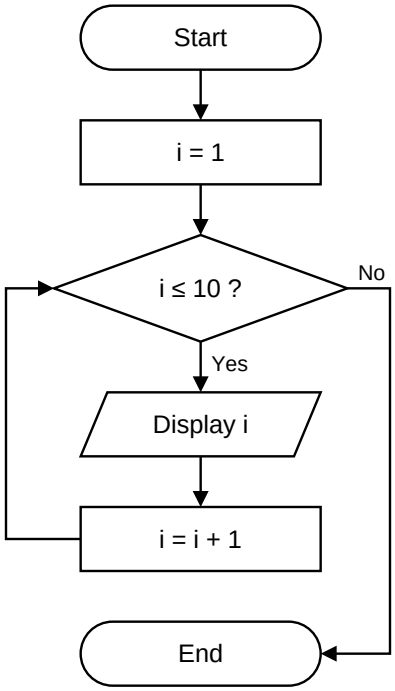
i	i ≤ 5 ?	Output
1	True	Maria
2	True	Maria
3	True	Maria
4	True	Maria
5	True	Maria
6	False	exit loop

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

2 Numbers from 1 to 10

Repetition structure: FOR

Display the numbers from 1 to 10.



Simulation table

Each row = one check of i ≤ 10; shaded row = loop ends.

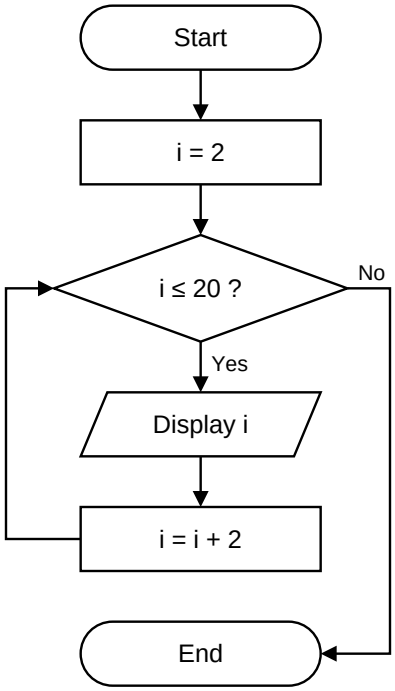
i	i ≤ 10 ?	Output
1	True	1
2	True	2
3	True	3
4	True	4
5	True	5
6	True	6
7	True	7
8	True	8
9	True	9
10	True	10
11	False	exit loop

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

3 Even Numbers

Repetition structure: FOR

Display all even numbers from 2 to 20.



Simulation table

Each row = one check of i ≤ 20; shaded row = loop ends.

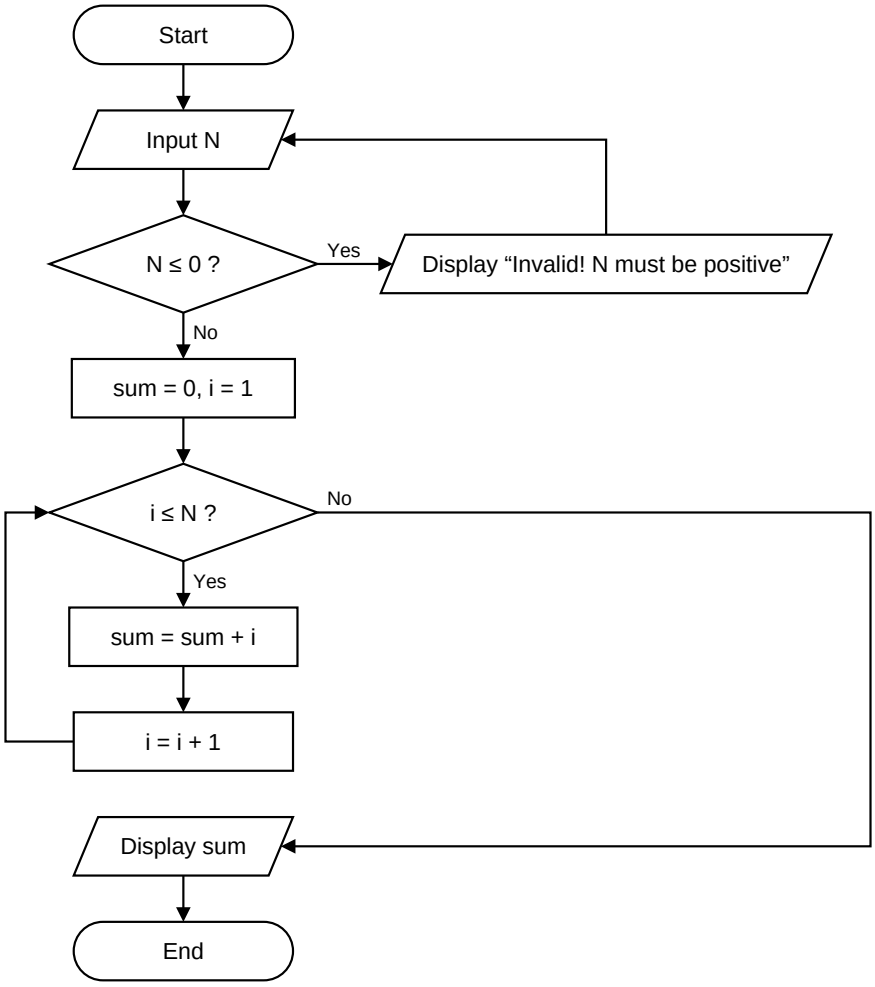
i	i ≤ 20 ?	Output
2	True	2
4	True	4
6	True	6
8	True	8
10	True	10
12	True	12
14	True	14
16	True	16
18	True	18
20	True	20
22	False	exit loop

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

4 Sum of Numbers

Repetition structure: DO-WHILE (input check) + FOR

Input a positive integer N. Compute and display the sum of all integers from 1 to N.



A. Input check

Inputs typed: −3, then 5. Shaded row = valid input, loop ends.

Attempt	N entered	N ≤ 0 ?	Action
1	−3	True	Display error, ask again
2	5	False	Accept N = 5

B. Sum loop

N = 5. Each row = one check of i ≤ N; shaded row = loop ends.

#	i	i ≤ N ?	sum (after)
init	1	—	0
1	1	True	1
2	2	True	3
3	3	True	6
4	4	True	10
5	5	True	15
6	6	False	15

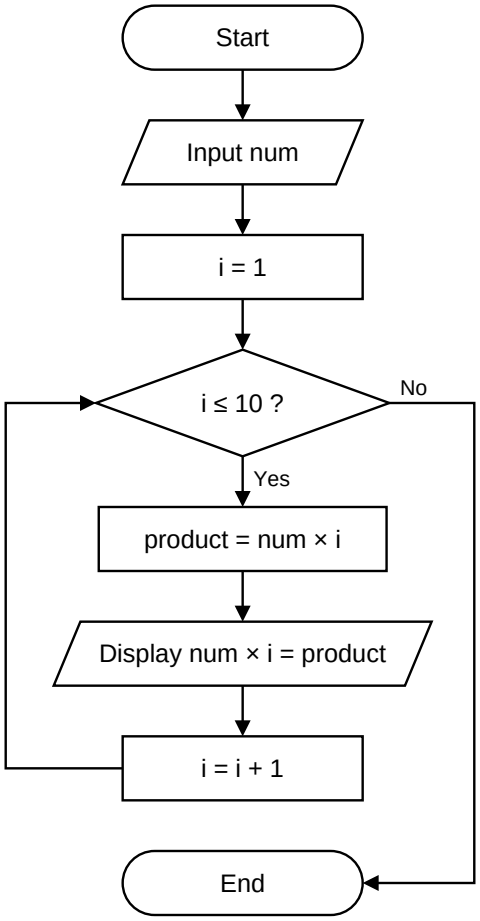
Output: Display sum → 15

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

5 Multiplication Table

Repetition structure: FOR

Input a number and display its multiplication table from 1 to 10.



Simulation table

Input: num = 7. Each row = one check of i ≤ 10; shaded row = loop ends.

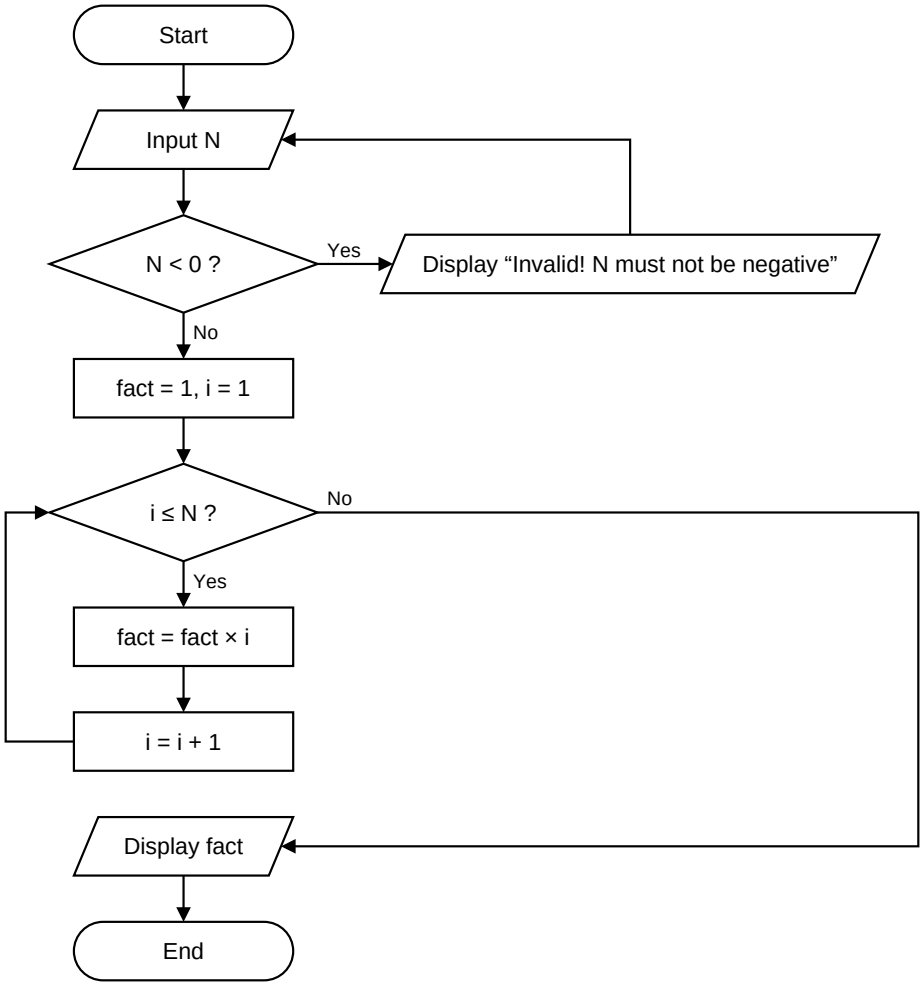
i	i ≤ 10 ?	product	Output
1	True	7	7 × 1 = 7
2	True	14	7 × 2 = 14
3	True	21	7 × 3 = 21
4	True	28	7 × 4 = 28
5	True	35	7 × 5 = 35
6	True	42	7 × 6 = 42
7	True	49	7 × 7 = 49
8	True	56	7 × 8 = 56
9	True	63	7 × 9 = 63
10	True	70	7 × 10 = 70
11	False	—	exit loop

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

6 Factorial of a Number

Repetition structure: DO-WHILE (input check) + FOR

Input a nonnegative integer N. Compute and display its factorial. N! = 1 × 2 × 3 × … × N



A. Input check

Inputs typed: −2, then 5. Shaded row = valid input, loop ends.

Attempt	N entered	N < 0 ?	Action
1	−2	True	Display error, ask again
2	5	False	Accept N = 5

B. Factorial loop

N = 5. Each row = one check of i ≤ N; shaded row = loop ends.

#	i	i ≤ N ?	fact (after)
init	1	—	1
1	1	True	1
2	2	True	2
3	3	True	6
4	4	True	24
5	5	True	120
6	6	False	120

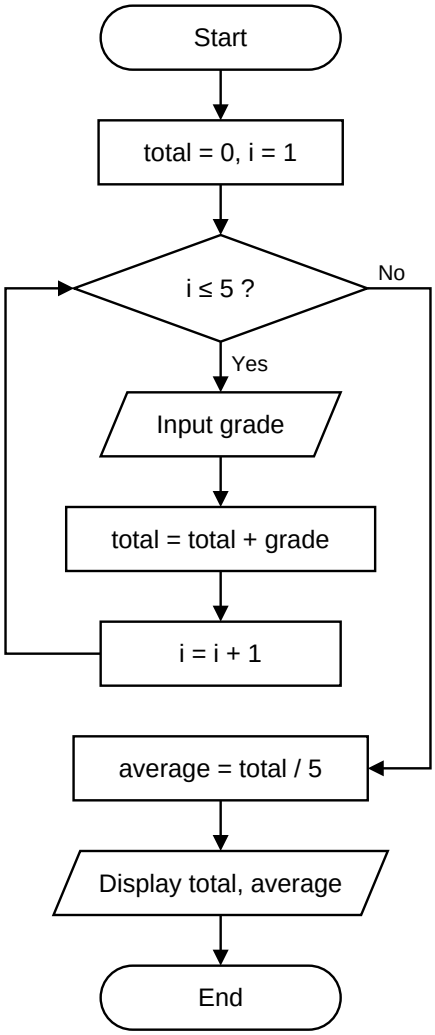
Output: Display fact → 120 (5! = 1 × 2 × 3 × 4 × 5)

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

7 Grade Average

Repetition structure: FOR

Input five grades. Compute and display their total and average.



Simulation table

Grades typed: 85, 90, 78, 92, 88. Shaded row = loop ends.

#	i	i ≤ 5 ?	grade	total (after)
init	1	—	—	0
1	1	True	85	85
2	2	True	90	175
3	3	True	78	253
4	4	True	92	345
5	5	True	88	433
6	6	False	—	433

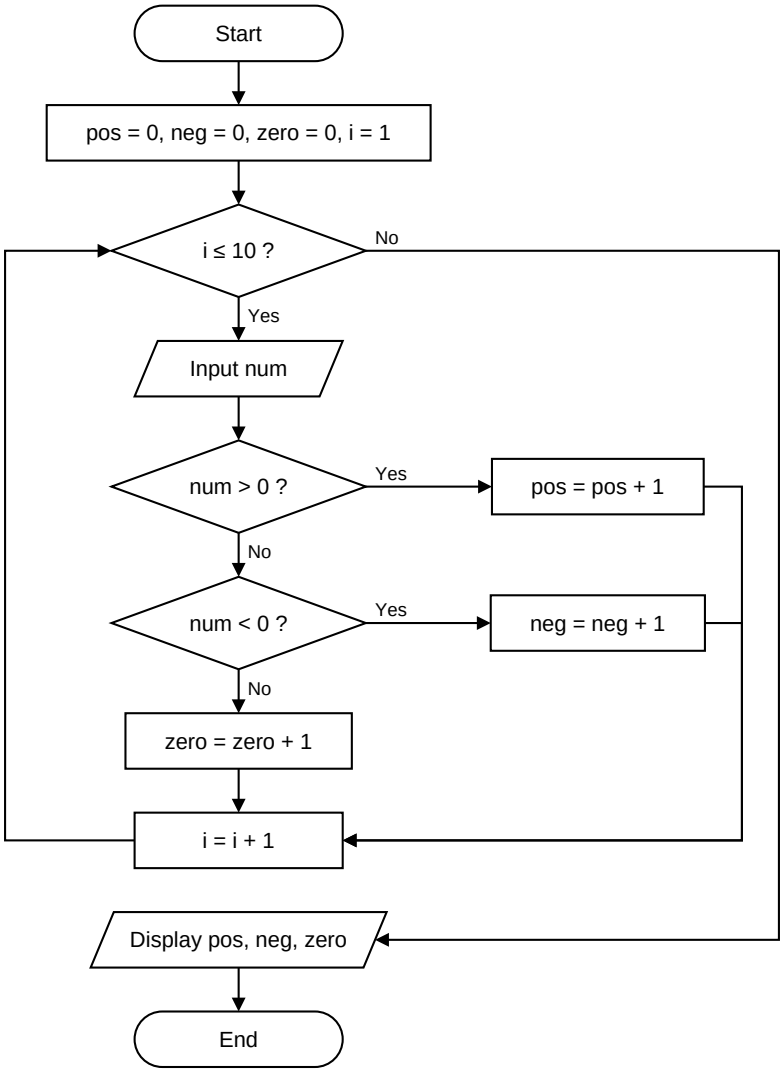
average = 433 / 5 = 86.6 **Output: Total = 433, Average = 86.6**

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

8 Count Positive and Negative Numbers

Repetition structure: FOR

Input ten numbers. Count and display how many are positive, negative, and zero.



Simulation table

Numbers typed: 5, −3, 0, 8, −7, 2, 0, −1, 9, 4. Shaded row = loop ends.

#	i	num	Branch	pos	neg	zero
init	1	—	—	0	0	0
1	1	5	positive	1	0	0
2	2	−3	negative	1	1	0
3	3	0	zero	1	1	1
4	4	8	positive	2	1	1
5	5	−7	negative	2	2	1
6	6	2	positive	3	2	1
7	7	0	zero	3	2	2
8	8	−1	negative	3	3	2
9	9	9	positive	4	3	2
10	10	4	positive	5	3	2
11	11	—	exit loop	5	3	2

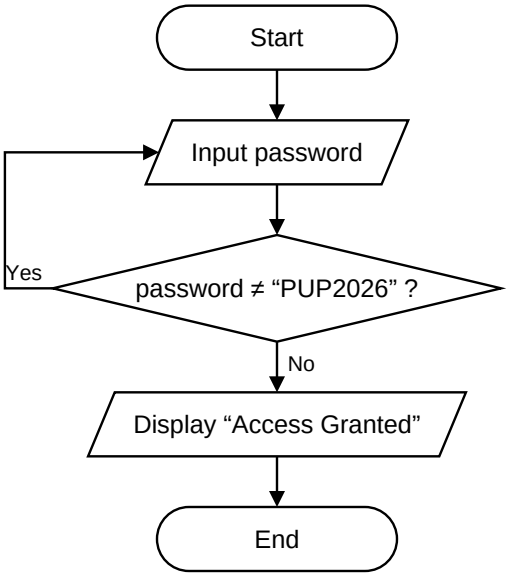
Output: Positive = 5, Negative = 3, Zero = 2

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution

9 Password Verification

Repetition structure: DO-WHILE

Ask the user to enter a password repeatedly until the correct password, PUP2026, is entered. Display “Access Granted” afterward.



Simulation table

Passwords typed: pup123, PUP2025, PUP2026. Shaded row = loop ends.

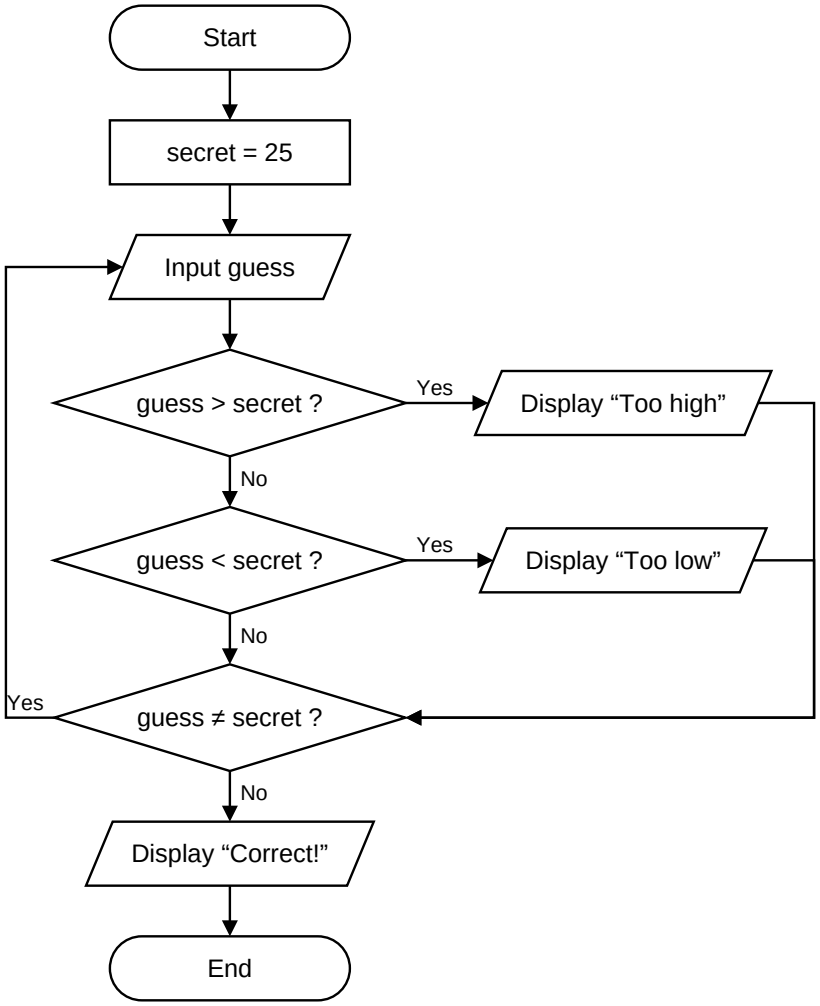
Attempt	password	password ≠ “PUP2026” ?	Action
1	pup123	True	Ask again
2	PUP2025	True	Ask again
3	PUP2026	False	Exit loop

Output: Display “Access Granted”

10 Number-Guessing Activity

Repetition structure: DO-WHILE

Set the secret number to 25. Repeatedly ask the user to enter a number until it matches the secret number. Display whether each incorrect guess is too high or too low.



Simulation table

secret = 25. Guesses typed: 40, 10, 30, 22, 25. Shaded row = loop ends.

Attempt	guess	Output	guess ≠ secret ?
1	40	Too high	True
2	10	Too low	True
3	30	Too high	True
4	22	Too low	True
5	25	—	False

Output: Display “Correct!” (the loop ends when guess = 25)

Start / End Input / Output Initialization, computation, counter update Loop condition Direction of execution